Photosynthesis

Photosynthesis is the biological process by which green plants, algae, and certain bacteria convert light energy into chemical energy stored in glucose. This process is fundamental to life on Earth as it produces oxygen and forms the base of most food chains.

The process occurs primarily in the chloroplasts of plant cells, specifically using a green pigment called chlorophyll that absorbs light, predominantly in the red and blue wavelengths of the visible spectrum.

Photosynthesis can be divided into two main stages: the light-dependent reactions, which occur in the thylakoid membranes and produce ATP and NADPH, and the light-independent reactions, also called the Calvin cycle, which occur in the stroma and use those products to synthesize glucose from carbon dioxide.

The overall chemical equation for photosynthesis is: six molecules of carbon dioxide plus six molecules of water, in the presence of light energy, produce one molecule of glucose plus six molecules of oxygen.

Photosynthesis plays a critical role in the global carbon cycle by removing carbon dioxide from the atmosphere, and it is the primary source of atmospheric oxygen that most living organisms depend on for respiration.